

Subtraction Is Conditional: An Interaction-Focused Replication of Deletion-First Refactoring Prompts

The Subtraction Research Project

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Abstract

Agentic coding systems often produce additive patches even when a task is framed as a refactor. We tested whether a previously observed deletion-first component effect would replicate across two larger deterministic refactor fixtures. The registered design crossed two tasks, three model/effort pairs, eight combinations of task-type, delete-first, and semantic net-line-budget components, and five repetitions: 240 planned cells. All 240 selected completed receipts passed the behavior oracle. The deletion-first factor was negative on the shared-normalizer fixture for GPT-5.6 Luna and Grok 4.5, but null for Composer; it was null or positive on the dead-compatibility-path fixture for all three models. The result is therefore task- and model-conditional subtraction, not a universal standalone D component. Six initial adapter failures are retained as separate transport provenance and excluded from completed comparisons. No priced cost claim is made.

1 Question and boundary

This study asks whether prompt components can shift the shape of an agentic refactor toward subtraction without sacrificing behavior. The claim tested here is deliberately narrow: does the deletion-first component produce a stable raw-net effect under a larger-refactor replication? A negative raw-net value means fewer net lines in the resulting patch; it is not, by itself, evidence of better engineering or lower cost.

2 Registered design

The wave used two deterministic tasks:

- refactor-shared-normalizer;
- refactor-dead-compatibility-path.

Each was crossed with the full eight-arm T/D/B factorial, three frozen model/effort pairs, and repetitions 1–5. Thus the planned size was

$$2 \times 3 \times 8 \times 5 = 240.$$

The models were GPT-5.6 Luna (maximum), Grok 4.5 (high), and Composer 2.5 (default). Completed comparisons require matched repetition IDs, completed adapter status, and passing behavior oracles. Initial failures remain visible in provenance artifacts rather than being silently treated as missing success.

3 Results

Table 1 reports the factorial D main effect and the matched D-only treatment-minus-neutral contrast, in raw-net lines. These are descriptive means over the five repetitions.

Table 1: Deletion-first effects by task and model. Negative values are more subtractive.

Task	Model	D factorial	D-only contrast
Shared normalizer	Composer 2.5	-0.65	0.0
Shared normalizer	GPT-5.6 Luna	-3.10	-6.4
Shared normalizer	Grok 4.5	-3.25	-10.0
Dead compatibility path	Composer 2.5	0.0	0.0
Dead compatibility path	GPT-5.6 Luna	+0.30	+0.8
Dead compatibility path	Grok 4.5	+2.0	+3.4

All selected observations passed their behavior checks: 240 of 240. The negative effect replicated across two models on one fixture, but not across both fixtures. This pattern is consistent with an interaction between task structure and model behavior. It does not establish a portable one-component mechanism, statistical significance, causality, or universality across repositories and maintenance tasks.

4 Provenance and cost boundary

The live run recorded six initial adapter failures. They were retried from fresh isolated roots and remain separate provenance; the completed dataset contains 240 receipts and excludes those six non-completed attempts from completed-run statistics. Token fields are preserved as descriptive telemetry. The planned priced multi-turn cost wave was deferred because the patch-shape effect did not survive both larger fixtures. No dollar-saving claim follows.

5 Conclusion

Deletion-first/reference-proof prompting can produce materially more subtractive patches on some larger refactor structures. In this replication, the effect is conditional on task and model. The next evidence gate is an independently specified refactor replication on a new task or repository before making broader claims or modeling priced trajectories.

Reproducibility

The protocol, findings, frozen manifest, receipts, factor effects, summary, and wave metadata are committed in the repository:

- `../research/phase-2/LARGER_REFACTOR_PROTOCOL.md`
- `../research/phase-2/LARGER_REFACTOR_FINDINGS.md`
- `../research/phase-2/design-larger-refactor-screen-r5.json`
- `../research/phase-2/live/larger-refactor-r5-results.json`

Build this paper from the repository root with Tectonic using the `paper/subtraction-study.tex` source, or from the `paper/` directory with `latexmk -pdf subtraction-study.tex`.